Key features of parabolas

1

a 5 units **b** 5 units **c** 10 units **d** 10 units

2

a 5 units **b** 5 units **c** 10 units **d** 10 units

3

a

i Upwards **ii** $(2, 4)$ **iii** $a = 3$ **iv** $(2, 7)$

v $y = 1$

b

i Upwards **ii** $(-4, -2)$ **iii** $a = \frac{3}{2}$ **iv** $\left(-4, -\frac{1}{2}\right)$

v $y = -\frac{7}{2}$

c

i Downwards **ii** $(3, -2)$ **iii** $a = 4$ **iv** $(3, -6)$

v $y = 2$

d

i Upwards **ii** $(0, 3)$ **iii** $a = \frac{1}{4}$ **iv** $\left(0, \frac{13}{4}\right)$

v $y = \frac{11}{4}$

e

i Upwards **ii** $(0, 0)$ **iii** $a = 4$ **iv** $(0, 4)$

v $y = -4$

f

i Upwards **ii** $(-2, 0)$ **iii** $a = 4$ **iv** $(-2, 4)$

v $y = -4$

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4

a

i To the right ii $(0,3)$ iii $a=3$ iv $(3,3)$

v $x=-3$

b

i To the right ii $(2,0)$ iii $a=4$ iv $(6,0)$

v $x=-2$

c

i To the right ii $(2,3)$ iii $a=4$ iv $(6,3)$

v $x=-2$

d

i To the left ii $(2,-4)$ iii $a=3$ iv $(-1,-4)$

v $x=5$

5

a

i $(0,0)$ ii $x=0$ iii $a=4$ iv $(0,4)$

v $y=-4$

b

i $(0,0)$ ii $x=0$ iii $a=\frac{1}{12}$ iv $\left(0,-\frac{1}{12}\right)$

v $y=\frac{1}{12}$

c

i $(0,0)$ ii $y=0$ iii $a=5$ iv $(-5,0)$

v $x=5$

d

i $(0,0)$ ii $y=0$ iii $a=\frac{1}{20}$ iv $\left(-\frac{1}{20},0\right)$

v $x=\frac{1}{20}$

6

a $(2,-1)$ **b** $(2,1)$ **c** $y=-3$ **d** $x=2$

e $(-\infty,\infty)$ **f** $[-1,\infty)$

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7

a $(-1, 3)$ **b** $(1, 3)$ **c** $x = -3$ **d** $y = 3$

e $[-1, \infty)$ **f** $(-\infty, \infty)$

8

a $x = \pm 2\sqrt{6}$ **b** $(2\sqrt{6}, -7)$ and $(-2\sqrt{6}, -7)$

9

a $y = \pm 2\sqrt{15}$ **b** $(-7, 2\sqrt{15})$ and $(-7, -2\sqrt{15})$

10 $k = -2,\ -6$

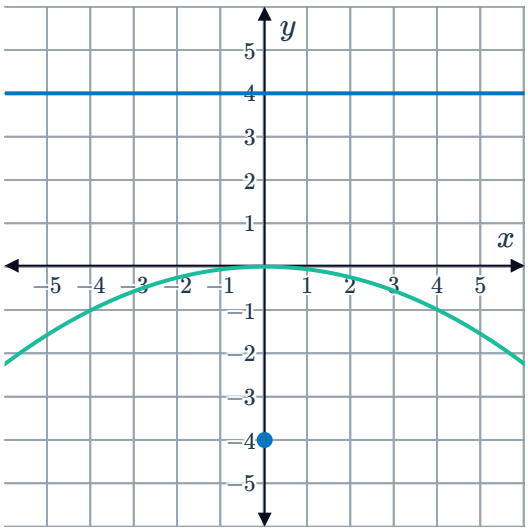
Graphs of parabolas

11

a Downwards **b** $(0, 0)$

c $a = 4$ **d** $(0, -4)$

e $y = 4$ **f**



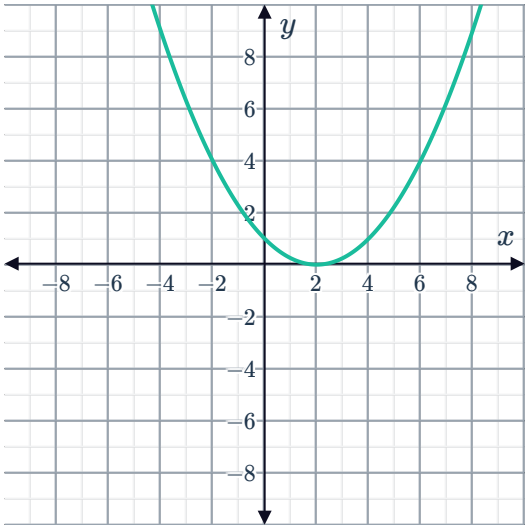
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12

a $(0, 1)$

b $y = -1$

c



d $(x - 2)^2 = 4y$

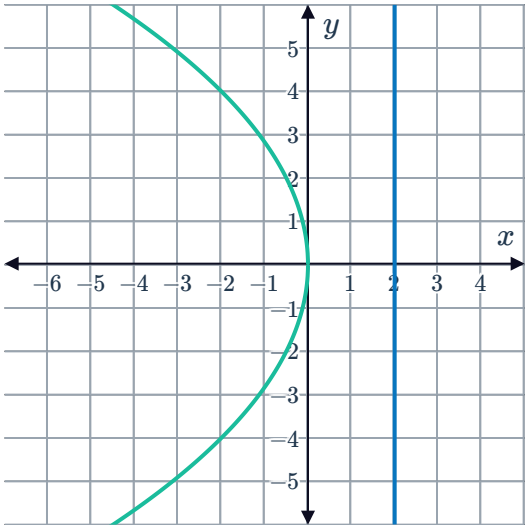
e $(2, 1)$

f $y = -1$

13

a $x = 2$

b



c $y = \pm 4$

d $(-2, 4)$ and $(-2, -4)$

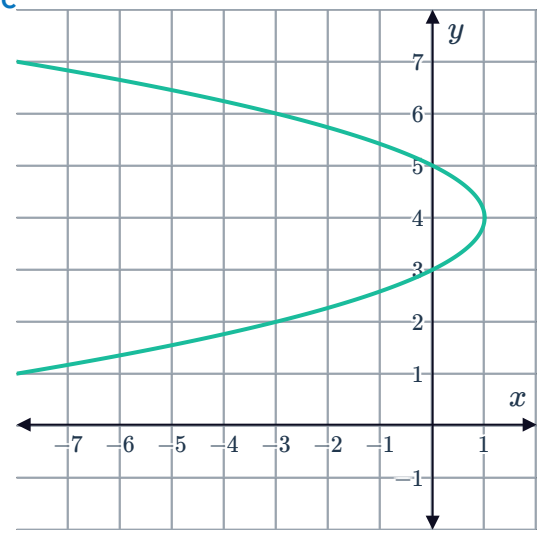
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14

a To the left.

b $(1, 4)$

c



15

a Downwards

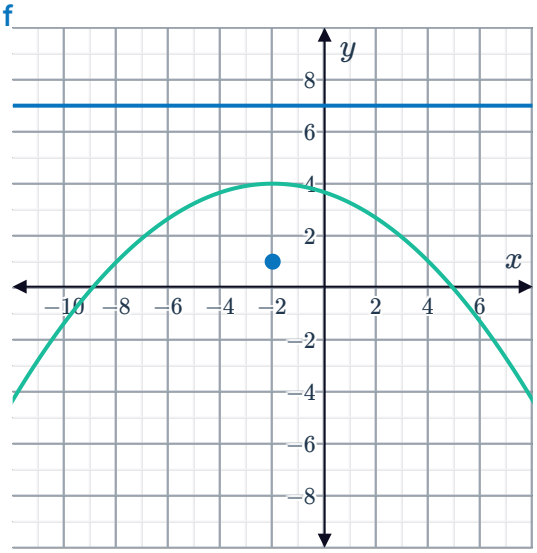
b $(-2, 4)$

c $a = 3$

d $(-2, 1)$

e $y = 7$

f



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16

a Upwards

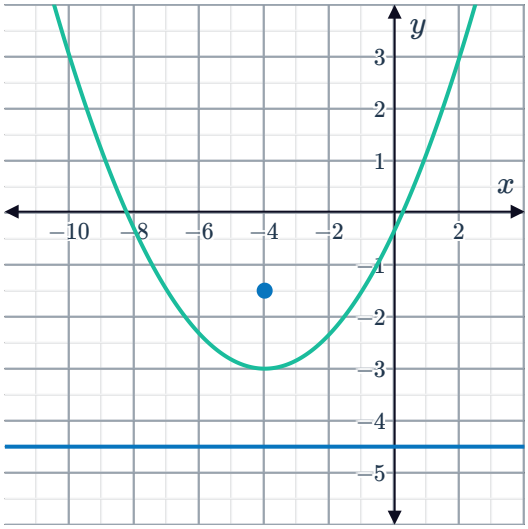
b $(-4, -3)$

c $a = \frac{3}{2}$

d $\left(-4, -\frac{3}{2}\right)$

e $y = -\frac{9}{2}$

f



g $\frac{15}{2}$ units

Equations of parabolas

17

a $x^2 = 16y$

b $x^2 = 12y$

c $y^2 = -24x$

d $y^2 = -2x$

e $x^2 = 2y$

f $x^2 = \frac{49}{5}y$

g $(x-6)^2 = 16y$

h $(y-8)^2 = 20x$

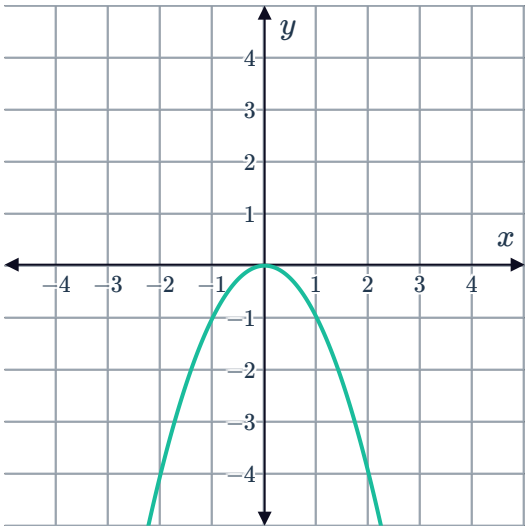
18 $y^2 = \pm 16x$

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19

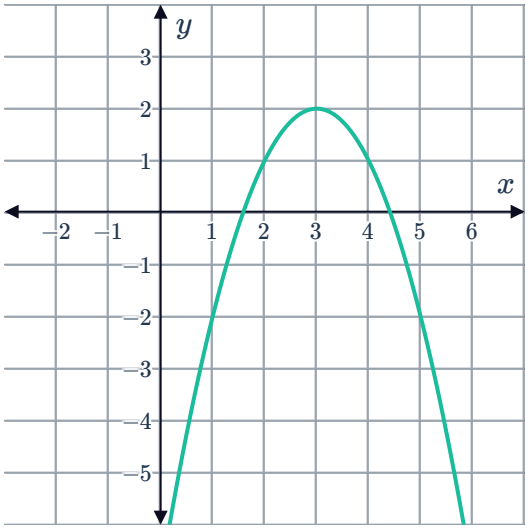
a $x^2 = -y$

b



20

a



b $(x-3)^2 = -(y-2)$

Applications

21

a $s = 12$

b 337

22

a $t = 3$

b 122

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